ONE-PAGE BRIEF

Post-quantum cryptography review: monero @ 3fd0f88c20f5

Client: Monero — public repository (passive review) · Source: github.com/monero-project/monero, commit `3fd0f88c20f5` · Reviewed: 2026-11-04 19:04 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **1** High **40** Medium
1 Low **1** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Key establishment: 10 + 5 − 8 = 7 → **too late**
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

High QSI-2026-12234	**Possible hard-coded secret in utils/python-rpc/framework/wallet.py** preliminary What to do: Move secrets to a secrets manager/environment; rotate the exposed values.
Medium QSI-2026-12236	**Transaction signing for 1 chain(s) relies on elliptic-curve signatures** What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium QSI-2026-12200	**Quantum-vulnerable Ed25519 key (Ed25519) in utils/gpg_keys/0xfffc0000.asc** What to do: Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it..
Medium QSI-2026-12201	**Quantum-vulnerable RSA key (4096-bit) in utils/gpg_keys/SNeedlewoods.asc** What to do: Inventory and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it..

38 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium channel key exchange (TLS, Noise, handshakes)	RSA	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)
Medium transaction and message signing	Ed25519	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	Ed25519, ECDH, X25519	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 1; unlikely to apply to this build: 1 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 0.

Static review of source code and manifests; files analysed: 1117. No requests were sent to the owner's systems.

Full report, content SHA-256: `474b1294ee360010a360896aa943fa57c4538bdf5d05a1386be0965270613489`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.